

ARK-459 — Results

Cloud IAM Role Grant · Revocation At Execution

Experiment ID: ARK-459
Institution: Remnant Fieldworks Inc.
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Substrate: Classical software (no quantum hardware, no cryptography)
Guards: Dual independent — V1 (JavaScript), V2 (Python)
Official execution: 2026-07-18 (post-lock — see `MANIFEST.txt`)

Verdict:  **PASS**

All four preregistered criteria met with perfect scores. When IAM role-grant authority is revoked before execution, the gate correctly **fails closed** (DENY when effective) or **fails safe** (HOLD when in-flight), never silently proceeding as if the authority were still valid.

Criterion	Requirement	Value	Result
C1 g_allow_valid_min	≥ 0.95	1.0000	 PASS
C2 g_deny_revoked_min	≥ 0.95	1.0000	 PASS
C3 g_hold_inflight_min	≥ 0.95	1.0000	 PASS
C4 g_margin	≥ 0.90	0.9500	 PASS

- Dual-guard concordance: **800/800 = 100.00%**
- Kill-gate negative control: wrongly ALLOWs **125/125 = 100%** of DENY/HOLD arms (testbed falsifiable ✓)

Per-Arm Results

Arm	Label	Expect	Correct	V1↔V2 Concordance
1	VALID-through-out	ALLOW	100/100	100/100
2	REVOKED-before-approval	DENY	100/100	100/100
3	REVOKED-after-approval-before-execution	DENY	100/100	100/100
4	REVOKED-during-multistep	DENY	100/100	100/100
5	IN-FLIGHT-at-execution	HOLD	100/100	100/100
6	REVOKED-then-REAUTHORIZED	ALLOW	100/100	100/100
7	IN-FLIGHT-boundary	HOLD	100/100	100/100
8	REVOKED-after-execution	ALLOW	100/100	100/100

Total: 800 decisions across 8 arms. All arms achieved **100% correctness** and **100% dual-guard concordance**.

Findings

1. **Perfect fail-closed and fail-safe behavior.** Arms 2–4 (revoked before execution, fully effective) all DENY 100/100. Arms 5, 7 (revocation in-flight at execution) all HOLD 100/100. No scenario in any DENY/HOLD arm was wrongly ALLOWed.
2. **Reauthorization handled correctly.** Arm 6 (revoked, then a new valid approval issued before execution) correctly ALLOWs 100/100 — the new approval governs, not the revoked one.
3. **“Revoked after execution” is not a failure.** Arm 8 (revocation issued strictly after `t_execution`) correctly ALLOWs 100/100 — the grant already executed under valid authority; the revocation is too late to affect it.
4. **Independent dual guards in perfect agreement.** V1 (JavaScript) and V2 (Python) produced identical decisions on all 800 scenarios — the logic is deterministic and the implementations are concordant.

5. **Anti-no-op oracle prevented defects.** Every generated scenario was validated by `_revocation_effective_oracle` before any guard saw it, ensuring the timing relationships were genuine. No scenario was mathematically inert (e.g., a “revoked-before-execution” arm whose revocation actually landed after execution).

Integrity / lock discipline

- The six locked files (`PREREGISTRATION.md` , both guards, generator, both run scripts) were hashed into `MANIFEST.txt` **before** the official run; post-run `./compute_hashes.sh` reproduced the identical hashes — the locked files were unchanged by execution.
- Both guards and the harness ran clean on first execution; no post-lock harness fix was required.

Honest bounds (unchanged from preregistration)

- Classical software test of **authorization control logic**, not a test of AWS, not a cryptographic security proof, not a production-readiness certification.
 - This tests the **timeline/revocation logic** in the IAM context; ARK-458 already tested exact-action-binding.
 - Results are bounded to this scenario model, these arms, and this seed family.
 - **No claim** that this experiment legally validates any patent claim or certifies RF-100 conformance.
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Published under the Remnant Fieldworks Standing Covenant (preregister → lock → execute → publish all outcomes). To God be the glory. Proof Before Power. Verification Before Execution.